PROJECT PROPOSAL: LYRICGPT

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1. Project Idea

Lyric generation is a challenging task in Natural Language Processing (NLP) that requires balancing creativity with linguistic coherence while incorporating musical characteristics such as rhythm, melody, tempo, and prosody. Traditional text generation models, including GPT-based architectures, primarily focus on linguistic patterns and often fail to account for the acoustic elements that influence lyrical flow and musicality.

This project proposes **LyricGPT**, a multi-modal Transformer-based lyric generation system that conditions text generation on both **textual** and **acoustic features**. By integrating audio-derived information such as tempo, pitch, beat structure, and key signatures into the generation process, the model aims to produce lyrics that are not only linguistically coherent but also musically aligned with the input audio.

The final objective is to build an AI-powered lyric generation framework capable of generating contextually relevant and rhythmically appropriate lyrics for a given song or audio sample.

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2. Plan of Action

The project will be developed through multiple phases. The timeline is approximate and allows flexibility for experimentation and refinement.

Phase 1: Research and Ideation (1–2 Weeks)

Goal

Develop a comprehensive understanding of lyric generation, multi-modal learning, and audio feature extraction techniques.

Tasks

- Review existing lyric generation approaches:
  - GPT-based models
  - LSTM-based architectures
  - Markov Chain methods
- Study audio feature extraction techniques using:
  - Librosa
  - Spotify API
  - Wav2Vec2
  - Essentia
- Explore multi-modal learning strategies:
  - Early fusion methods
  - Late fusion methods
  - Cross-modal attention mechanisms
- Identify suitable datasets containing lyrics and corresponding audio.
- Set up the development environment and select frameworks:
  - PyTorch
  - TensorFlow
  - Hugging Face Transformers

Deliverable

A detailed literature review and finalized technical stack.

Phase 2: Data Collection and Preprocessing (2–3 Weeks)

Goal

Construct a high-quality dataset containing aligned lyrics and audio features.

Tasks

Lyric Collection

- Gather lyrics from:
  - Musixmatch
  - Genius
  - Kaggle lyric datasets
- Clean and preprocess textual data.
- Tokenize lyrics and remove inconsistencies.

Audio Feature Extraction

Extract musical attributes including:

- Tempo
- Beat positions
- Key signatures
- Pitch information
- Spectral characteristics

using tools such as:

- Librosa
- Essentia
- Spotify API

Text-Audio Alignment

- Pair lyric segments with corresponding audio embeddings.
- Normalize extracted features.
- Remove noisy or incomplete samples.
- Apply data augmentation techniques:
  - Tempo shifts
  - Pitch transpositions
  - Audio perturbations

Deliverable

A structured dataset consisting of:

(Lyrics, Audio Features, Metadata)

Phase 3: Model Design and Architecture Selection (3–4 Weeks)

Goal

Develop and evaluate a multi-modal Transformer architecture for lyric generation.

Tasks

Base Model Selection

Evaluate and compare:

- GPT-2
- T5
- BART
- LLaMA

Multi-Modal Fusion Strategies

Experiment with:

1. **Early Fusion**
    - Concatenation of audio features with text embeddings.
2. **Cross-Attention Mechanisms**
    - Conditioning text generation on audio representations.
3. **Dual Encoder Architecture**
    - Separate encoders for text and audio modalities.

Rhythmic Alignment

- Modify positional encodings to better capture rhythmic and temporal information.
- Incorporate beat-aware embeddings where applicable.

Prototype Development

- Train an initial model using a subset of the dataset.
- Evaluate feasibility and training stability.

Deliverable

A preliminary multi-modal lyric generation architecture with initial experimental results.

Phase 4: Training and Optimization (4–6 Weeks)

Goal

Train and optimize the complete lyric generation model.

Tasks

- Train the model using combined textual and acoustic inputs.
- Fine-tune pre-trained language models on lyric-specific datasets.
- Design a loss function that balances:

- Text coherence
    - Semantic relevance
    - Rhythmic consistency
- Perform hyperparameter tuning:
    - Learning rate
    - Batch size
    - Dropout rate
    - Number of layers
    - Attention heads
- Monitor performance and prevent overfitting.

Deliverable

A fully trained LyricGPT model capable of generating musically aligned lyrics.

Phase 5: Evaluation and Benchmarking (3–4 Weeks)

Goal

Assess the quality and effectiveness of generated lyrics.

Evaluation Metrics

Text-Based Metrics

- BLEU Score
- ROUGE Score
- Perplexity

Music-Aware Metrics

- Syllable-to-beat alignment
- Rhythmic consistency
- Genre consistency
- Semantic relevance to song characteristics

Human Evaluation

Collect feedback from:

- Musicians
- Lyricists
- General listeners

Comparative Analysis

Compare different:

- Base architectures
- Fusion techniques
- Feature extraction methods

Deliverable

Identification of the best-performing architecture based on linguistic quality and musical alignment.

Phase 6: Deployment and Future Extensions (Final Phase)

Goal

Create an accessible interface for real-world usage.

Tasks

Develop a web-based application or API that allows users to:

- Upload an audio file or song sample.
- Generate lyrics aligned with the song's rhythm and style.
- Customize generation parameters such as:
  - Genre
  - Mood
  - Creativity level
  - Language style

Future Extensions

- Multilingual lyric generation
- Melody-conditioned lyric generation
- Real-time songwriting assistance
- Integration with music production software

Final Deliverable

A deployed AI-powered lyric generation system capable of producing rhythm-aware and contextually relevant lyrics from audio inputs.

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3. Expected Outcomes

The project is expected to:

- Demonstrate the effectiveness of multi-modal learning for lyric generation.
- Improve lyrical alignment with musical structure compared to text-only language models.
- Provide a practical framework for AI-assisted songwriting.
- Establish a foundation for future research in music-aware generative AI systems.

References

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